

PERIMETER SECURITY SYSTEM – OPERATING INSTRUCTIONS

Infrared Barriers, Microwave Sensors, Seismic Detectors Polish Border Guard – ISO-Compliant Version

Author: Aleksandra G. – Technical Writer / Operational Safety Documentation

Applicable Standards: ISO 20607, ISO 12100, ISO 13849-1, ISO 11064

Intended Users: Trained Border Guard officers assigned to the operator workstation

Document Type: Operating Instructions (Safety-Relevant)

1. Identification and Purpose

This document provides standardized operating instructions for the **Perimeter Security System** used by the Polish Border Guard.

The system may include:

- **Infrared (IR) barriers** – primary detection technology used in BG,
- **Microwave barriers** – optional supplementary detection,
- **Seismic sensors** – optional underground detection,
- **Video verification subsystem** – PTZ cameras automatically directed to alarm zones.

The purpose of the system is to **detect, verify, and respond to intrusion attempts** or **illegal border crossings**.

2. Intended Use

The system is intended to:

- detect unauthorized movement across the protected perimeter,
- support real-time monitoring from the operator workstation,
- enable rapid dispatch of patrol units,
- provide video evidence for operational and legal purposes.

Misuse to be avoided:

- disabling sensors without authorization,
- covering or obstructing detection devices,
- ignoring alarms without verification.

3. System Description

3.1. Infrared (IR) Barriers

- Consist of an IR transmitter and receiver.
- Alarm is triggered when the IR beam is interrupted.
- Typical range: 30–200 m.
- Installed at 0.8–1.2 m or in multi-beam configuration.

3.2. Microwave Barriers

- Create an electromagnetic detection field.
- Detect movement within the protected zone.
- Used in areas with uneven terrain or vegetation.

3.3. Seismic Sensors

- Installed underground.
- Detect ground vibrations caused by walking or vehicle movement.

3.4. Video Verification Subsystem

- Cameras automatically rotate to the alarm location.
- Operator performs real-time verification

4. Safety Information

4.1. General Safety Warnings

WARNING – Security-Critical System Unauthorized deactivation or manipulation may compromise national border security.

WARNING – Operational Responsibility Every alarm must be treated as a potential threat until verified.

4.2. Operator Safety Requirements

- The operator must remain at the workstation at all times during duty.
- Only trained personnel may operate the system.
- All malfunctions must be reported immediately.

4.3. Equipment Handling

- Do not cover IR transmitters or receivers.
- Do not direct flashlights or laser beams at sensors.
- Do not attempt repairs; only authorized technicians may service equipment.

5. Pre-Operation Checks

Before starting the monitoring shift, the operator must:

1. Verify that all sensors are online and communicating with the control panel.
2. Check that camera feeds are functioning.
3. Confirm that no zones are disabled without authorization.
4. Review the logbook for previous shift notes.

6. Normal Operation

6.1. Operator Responsibilities

The operator shall:

- continuously **monitor the alarm panel**,
- observe **live camera feeds**,
- perform **video verification** of each alarm,
- classify events according to Section 7,
- dispatch patrol units when required,
- document all alarms and actions.

CAUTION: The operator must not leave the workstation until relieved.

7. Alarm Handling Procedure

When an alarm is triggered:

Step 1 – Alarm Reception

Read the alarm details (zone, sensor type, timestamp).

Step 2 – Automatic Camera Positioning

The system automatically directs the PTZ camera to the alarm location.

Step 3 – Video Verification

The operator must immediately switch to the camera feed and classify the event:

7.1. Event Classification

A. Animal (fox, deer, dog)

Classification: *Natural false alarm*

Action: Log the event; no patrol required.

B. Environmental Conditions (fog, snow, moving branches)

Classification: *Environmental alarm*

Action: Log the event; notify technician if recurring.

C. Person or Vehicle Near the Barrier

Classification: *Suspicious activity / reconnaissance*

Action: Notify the duty officer.

- **Dispatch a patrol to the alarm zone.**

D. Person Crossing the Barrier Line

Classification: **ILLEGAL BORDER CROSSING / INTRUSION**

Action:

1. Immediately notify the duty officer.
2. **Dispatch a patrol without delay.**
3. Monitor the intruder's movement via camera.
4. Provide patrol with real-time updates (direction, number of individuals, behavior).
5. Secure video footage.

8. Video Verification Guidelines

The operator must:

- assess direction and speed of movement,
- determine number of individuals,
- evaluate whether the object is approaching or retreating,
- identify carried objects (tools, bags, equipment),
- treat ambiguous cases as **potential threats**.

9. Post-Alarm Actions

After handling the alarm:

- record the event in the system log,
- create a service note if required,
- secure video evidence (for intrusions),
- report technical issues,
- brief the next shift.

10. Maintenance and Inspection

Daily (Operator)

- Verify camera visibility and sensor status.
- Confirm alarm panel functionality.

Weekly (Technician)

- Perform range tests.
- Conduct alarm tests.
- Inspect mounting and alignment.

Monthly

- Full system diagnostics.
- Clean optical components.
- Apply software updates.

11. Decommissioning and Shutdown

System shutdown is permitted **only with written authorization**. All zones must be reactivated before the next shift.